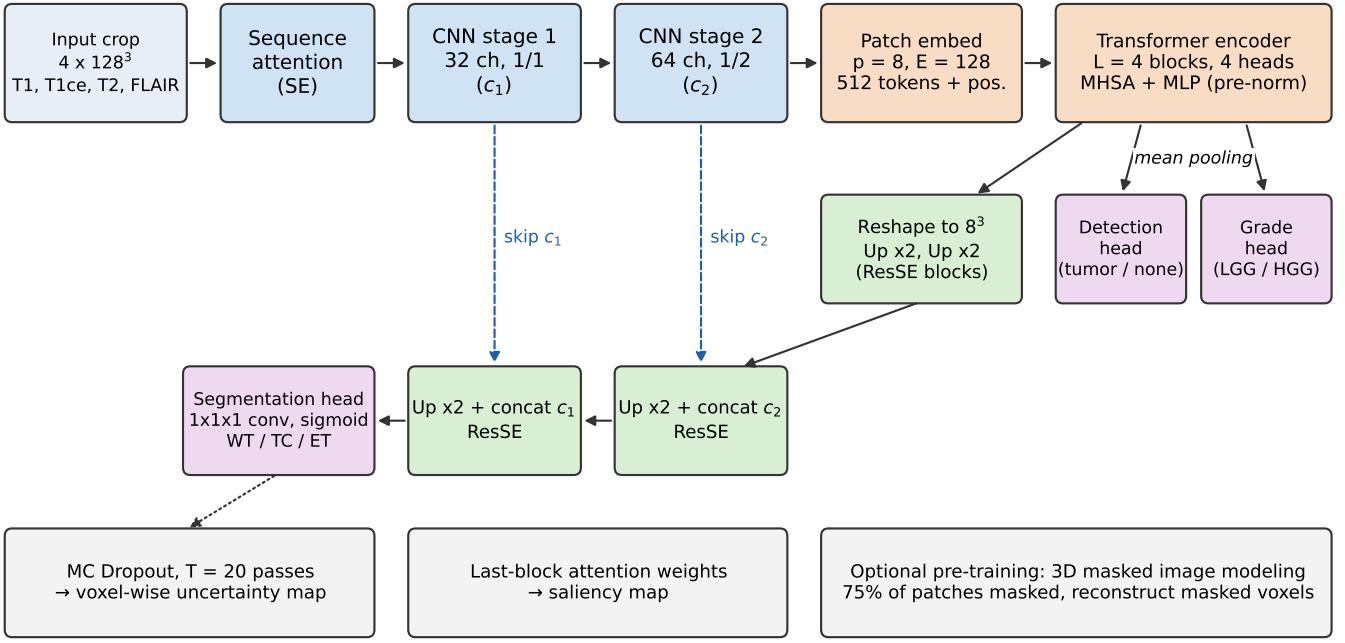
Input crop
4 x 128³
T1, T1ce, T2, FLAIR
Sequence attention (SE)
CNN stage 1
32 ch, 1/1
(c_1)
CNN stage 2
64 ch, 1/2
(c_2)
Patch embed
p = 8, E = 128
512 tokens + pos.
Transformer encoder
L = 4 blocks, 4 heads
MHSA + MLP (pre-norm)
mean pooling
skip c_1
skip c_2
Reshape to 8³
Up x2, Up x2
(ResSE blocks)
Detection head
(tumor / none)
Grade head
(LGG / HGG)
Segmentation head
1x1x1 conv, sigmoid
WT / TC / ET
Up x2 + concat c_1
ResSE
Up x2 + concat c_2
ResSE
MC Dropout, T = 20 passes
→ voxel-wise uncertainty map
Last-block attention weights
→ saliency map
Optional pre-training: 3D masked image modeling
75% of patches masked, reconstruct masked voxels